

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [219, 153, 156, 122, 119, 143, 130, 110, 120, 126]

Tap amplitudes (pixels): [120.7683733433551, 128.03515142334936, 125.39936203984452, 108.24047302187847, 93.77632963600144, 89.47066558375433, 100.4987562112089, 90.4267659490264, 92.17917335276988, 90.78546139112804, 110.57124400132251]

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Analysis of extracted features against MDS-UPDRS 3.4 criteria

Given data (10-tap sequence inferred from 11 tap peaks):

- Number of taps detected: 11 peaks → consistent with 10 intervals (suggests 10 intended taps plus start/end detection; we evaluate 10 consecutive taps as requested)
- Tap periods (frames): [219, 153, 156, 122, 119, 143, 130, 110, 120, 126]
 - Pattern: Rapid initial period then stabilizes; no clear long arrests. Variation present but no extreme outliers after the first interval.
 - Interruptions/hesitations: No obvious long pauses; the largest gap is the first (219), then all are between ~110–156. That suggests at most occasional brief hesitations, not 3–5 interruptions.
 - Speed: After the first interval, periods cluster ~110–156 frames and tend to shorten vs. the first, indicating slight-to-mild slowing overall but not moderate.
- Tap amplitudes (pixels): [120.8, 128.0, 125.4, 108.2, 93.8, 89.5, 100.5, 90.4, 92.2, 90.8, 110.6]
 - Amplitude trajectory: High for taps 1–3, then notable drop by tap 4–6, staying lower through taps 7–10, with a late rebound at the 11th detected peak.
 - Decrement timing: Clear decrease begins by tap 4 (i.e., “starting after the 1st tap,” occurring early rather than only near the end).

- Magnitude of decrement: From ~125 average early amplitude to ~90–100 mid/late, roughly 20–30% decrease, sustained across many taps.

Mapping to scoring criteria:

- Interruptions: Not “more than 5” and no clear long arrest → does not force a 3 on interruptions.
- Speed: Variability with some longer early period, but post-first-tap cadence is fairly consistent; this supports slight–mild slowing rather than moderate.
- Amplitude decrement:
 - Not just “near the end” (score 1c), and not merely “midway” (score 2c).
 - The decrement is apparent starting early (by tap ~4) and sustained, aligning best with “The amplitude decrements starting after the 1st tap” → criterion 3c (Moderate).

Given MDS-UPDRS scoring selects the highest applicable severity among the domains (speed, rhythm interruptions, amplitude), the early and sustained amplitude decrement fulfills the 3-point criterion even though interruptions and speed alone would score lower.

Final score: 3
